Z-Scores & Linear Transformations

ANSWER KEY

Section 1: Calculating and Interpreting Z-Scores

1 A student scores 82 on an exam where the class mean is 74 and the standard deviation is 5. What is the student's standardized z-score?

a. 1.60 ✓

b. 1.20

c. 0.80

d. -1.60

$z = \frac{x - \mu}{\sigma} = \frac{82 - 74}{5} = \frac{8}{5} = 1.60$.

2 What does a z-score of -2.4 indicate about an observation?

a. **The observation is 2.4 standard deviations below the mean** ✓

b. The observation is 2.4 units below the mean

c. The observation is in the 2.4th percentile

d. The probability of the observation is negative

A z-score measures the signed number of standard deviations an observation falls relative to the arithmetic mean; negative indicates below the mean.

3 On Test 1, Alice scored 88 (mean 80, SD 4). On Test 2, she scored 92 (mean 85, SD 5). On which test did she perform relatively better compared to peers?

a. **Test 1 ($z_1 = 2.0 > z_2 = 1.4$)** ✓

b. Test 2 ($z_2 = 1.4 > z_1 = 1.0$)

c. Both performances were identical

d. Test 2 because her raw score was 92 vs 88

Test 1: $z_1 = \frac{88 - 80}{4} = 2.0$. Test 2: $z_2 = \frac{92 - 85}{5} = 1.4$. Performance relative to peers is higher on Test 1.

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Section 2: Linear Transformations - Adding a Constant

4 A dataset has mean $\mu = 50$ and standard deviation $\sigma = 8$. If 5 points are added to every observation ($y = x + 5$), what are the new mean and standard deviation?

a. **New mean** $= 55$**, New** $\text{SD} = 8$ ✓

b. New mean $= 55$, New $\text{SD} = 13$

c. New mean $= 50$, New $\text{SD} = 13$

d. New mean $= 55$, New $\text{SD} = 40$

Adding a constant shifts all measures of center (mean increases by 5 to 55), but does not alter measures of spread (standard deviation remains 8).

5 If a constant $c = 10$ is subtracted from every value in a dataset, which statistic changes?

a. **Median** ✓

b. Range

c. IQR

d. Variance

Subtracting a constant shifts the center (median decreases by 10), but leaves all measures of spread (range, IQR, variance, standard deviation) unchanged.

6 A teacher curves a test by adding 8 points to all student scores. How does this affect the shape of the score distribution?

a. **The shape remains exactly the same** ✓

b. The distribution becomes more symmetric

c. The distribution becomes skewed to the right

d. The spread increases

Adding a constant is a rigid shift of the distribution along the axis; it has zero impact on the shape or spread.

Section 3: Linear Transformations - Multiplying by a Constant

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7 A dataset of temperatures in Celsius has mean 20°C and standard deviation 4°C. Converting to Fahrenheit using $F = 1.8C + 32$, what are the new mean and standard deviation?

a. **Mean $= 68^\circ\text{F}$, SD $= 7.2^\circ\text{F}$** ✓

b. Mean $= 68^\circ\text{F}$, SD $= 39.2^\circ\text{F}$

c. Mean $= 36^\circ\text{F}$, SD $= 7.2^\circ\text{F}$

d. Mean $= 68^\circ\text{F}$, SD $= 4.0^\circ\text{F}$

New mean: $1.8(20) + 32 = 36 + 32 = 68^\circ\text{F}$. New standard deviation: $|1.8| \times 4 = 7.2^\circ\text{F}$ (addition of $+32$ does not affect spread).

8 If all observations in a dataset with variance 16 are multiplied by -3, what is the new variance?

a. 144 ✓

b. -48

c. 48

d. 12

Multiplying by constant b multiplies variance by b^2. Here $(-3)^2 \times 16 = 9 \times 16 = 144$.

9 If every observation is multiplied by 0.5, how does the IQR change?

a. **The IQR is halved (multiplied by 0.5)** ✓

b. The IQR is unchanged

c. The IQR is quartered (multiplied by 0.25)

d. The IQR doubles

Measures of spread are multiplied by $|b|$. With $b = 0.5$, the new IQR is $0.5 \times$ old IQR.

Section 4: Standardizing an Entire Distribution

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10 If an entire non-normal distribution with mean $\mu = 40$ and SD $= 6$ is converted to z-scores ($z = \frac{x - 40}{6}$), what are the mean and standard deviation of the standardized values?

a. **Mean $= 0$, Standard Deviation $= 1$ ✓**

b. Mean $= 40$, Standard Deviation $= 1$

c. Mean $= 0$, Standard Deviation $= 6$

d. Mean $= 1$, Standard Deviation $= 0$

Standardizing any distribution (regardless of shape) by subtracting its mean and dividing by its standard deviation always produces a new distribution with mean 0 and standard deviation 1.

11 Does standardizing a skewed dataset into z-scores transform the distribution into a normal shape?

a. **No, standardizing changes center and scale but preserves the original skewed shape ✓**

b. Yes, standardizing always creates a standard normal bell curve

c. Only if the sample size is greater than 30

d. Yes, because the mean becomes 0 and standard deviation becomes 1

Standardization is a linear transformation ($z = \frac{1}{\sigma}x - \frac{\mu}{\sigma}$). Linear transformations never change the fundamental shape of a distribution.

12 A runner's race time is at the 85th percentile of all competitors. What does this mean?

a. **The runner was faster than 85% of competitors (finished with lower time than 15% of competitors) ✓**

b. The runner scored an 85% on the race

c. The runner finished in the top 85%

d. 85% of competitors finished before this runner

In general percentile terminology, the pth percentile is the value such that p percent of observations fall at or below it.

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